DIOGO SANTOS

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PROFESSIONAL EXPERIENCE

Systems Engineer - Cloudflare (*Lisbon, Portugal*) Mar 2026 - Present

- Working on Email Routing and Email Sending.

Junior Protocol Engineer - DV Labs (*Remote*) Nov 2024 - Feb 2026

- Worked on Charon, a middleware software that transforms a single-node service into a distributed cluster, improving fault tolerance, availability and security through consensus-based coordination and threshold signatures.
- Researched and implemented consensus protocols such as Raft to reduce consensus latency.
- Enhanced cluster observability by adding new Prometheus metrics and built Grafana dashboards and alerting rules to proactively monitor system health.
- Upgraded peer networking from TCP to QUIC, implemented HTTP request fallback logic, added smart contract signature verification, configured SSE event listeners for improved attestation timings among other features.

Skills: Golang, Docker, Prometheus, Loki, Grafana, Github Actions, CI

RESEARCH EXPERIENCE

First Stage Researcher - INESC-ID (*Lisbon, Portugal*) May 2024 - Jul 2024

- Implemented Alea-BFT, a leaderless, asynchronous consensus algorithm, to substitute QBFT in Charon, a DV Labs CLI for running distributed Ethereum validators.
- Demonstrated that Alea-BFT handled crash and delay faults up to 10-50x more effectively than QBFT, with significantly less performance degradation in adverse scenarios while maintaining equivalent latency, throughput, and resource usage, ensuring no trade-off in the "happy path."

Skills: Golang, Docker

Research Initiation Grant - Instituto Superior Técnico (*Lisbon, Portugal*) Sep 2022 - Apr 2024

- Built a full-stack web application to manage the Computer Science Department's dissertation defense scheduling process, reducing administrative manual labor.
- Collaborated weekly with professors and staff to gather requirements, run user tests, and iterate the UI.
- Built backend services to support bulk import/validation of thesis proposals, automated email notifications, jury availability tracking and conflict-free room reservation assignment, and automated generation of final documents with grades.

Skills: Vue.js (Typescript), Spring Boot (Java), MariaDB, Docker

INTERNSHIPS

Software Engineer Intern - Jungle AI (*Lisbon, Portugal*) Jul 2023 - Aug 2023

- Developed a parallel data processing workflow in Python utilizing the Ray framework alongside AWS services such as S3 and SQS and deployed a Ray cluster on Kubernetes.

Front-End Developer Intern - Spin.Works (*Lisbon, Portugal*) Jul 2021 - Sep 2021

- Developed a PWA in Angular.js with interactive map navigation (similar to Google Maps) enabling farmers to explore and manage different farming zones for precision agriculture.

EDUCATION

Master of Computer Science and Engineering, Instituto Superior Técnico 2022 - 2024
GPA: 18/20 — Academic Merit Diploma (2022/23), Academic Excellence Diploma (2023/24)
Bachelor of Computer Science and Engineering, Instituto Superior Técnico 2019 - 2022
GPA: 17/20 — Academic Merit Diploma (2019/20, 2020/21, 2021/22)